MIL-G-85468(AS)
7 August 1981

MILITARY SPECIFICATION

GENERATOR-CONTROL-INDICATOR
SG-1189/AST-4(V)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the requirements for the design, manufacture, inspection, and testing of the Generator-Control-Indicator SG-1189/AST-4(V), referred to hereafter as the generator.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-T-5422	Testing, Environmental, Airborne Electronic and Associated Equipment.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Air Engineering Center, Engineering Specifications and Standards Department (ESSD) Code 93, Lakehurst, New Jersey 08733, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5841

STANDARDS

MILITARY

DOD-STD-100	Engineering Drawing Practices.
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of U.S. Military Property.
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of.
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of.
DOD-STD-480	Configuration Control - Engineering Changes, Deviations and Waivers.
MIL-STD-704	Aircraft Electrical Power Characteristics.
MIL-STD-781	Reliability Design Qualification and Production Acceptance Test: Exponential Distribution.
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of.
MIL-STD-810	Environmental Test Methods.
MIL-STD-831	Test Reports, Preparation of.

DRAWINGS

NAVAL AIR SYSTEMS COMMAND
(Code Ident 30003)

DL 1236AS500	Generator-Control-Indicator SG-1189/AST-4(V).

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 First article. When specified (see 6.2.1) the contractor shall furnish sample units for first article inspection and approval (see 4.4 and 6.3).

3.1.1 Item description. The generator is a pulse generator used in airborne cockpit installations to control operation of the Transmitting Set, Radar AN/DPT-1(V). Both are subassemblies of the Pod, Simulating Set CY-7927/AST-4(V).

3.1.2 First article sample. Unless otherwise specified in the contract o purchase order (see 6.2.1), a first article sample of two generators shall delivered to the testing activity designated in the contract or purchase order.

The first article sample shall be manufactured using the same methods, materials, processes and procedures proposed for production. Any production prior to acceptance of the first article sample is at the contractor's risk.

3.2 Selection and quality verification of parts and materials. Quality verified parts (see 3.2.1) and materials from approved sources (see 3.2.2) shall be used in the fabrication of the end item. The contractor shall prepare test reports of conformance to applicable requirements.

3.2.1 Approval of parts and materials. Parts and materials proposed for use in the fabrication of the end item that are not specified in detail herein, or in applicable referenced documents, shall be selected from other documents in the order of precedence specified in MIL-STD-143, and shall be approved by the procuring activity prior to use. Approval shall be predicated upon source (vendor) approval and verification of part quality by test.

3.2.2 Source approval. The contractor shall be responsible for ascertaining that each proposed source for parts and materials maintains a quality control program to ensure consistent delivery of products conforming to applicable requirements. When parts or materials are fabricated by the contractor, the area used for that purpose shall be considered the same as an outside vendor insofar as source approval is concerned. Source approvals by the contractor shall be subject to concurrence by the procuring activity. Source approval will be granted by the procuring activity upon preparation of acceptable evidence of any of the following:

a. Formal on-site survey of the vendor's manufacturing facilities.

b. The source has been previously approved, and evidence is available to confirm that acceptable product quality has been maintained.

c. Other evidence of acceptable product quality is available.

3.2.3 Verification of part quality. Quality verification shall be demonstrated by successful testing of the part to levels of environmental, electrical, and physical stresses equal to or greater than those which will be encountered in the system and as required by the part specification. Part quality may be demonstrated by any of the following:

a. When the part is manufactured and qualified to a military specification which includes requirements equal to or more severe than those of the end-item application.

b. When the part is manufactured and qualified to a military specification not including all requirements of the application, and additional tests for such requirements have been performed.

c. When the part is not qualified to a military specification, but all tests appropriate to the application have been performed.

3.2.4 Approved source/quality verified parts list (AS/QVPL). An AS/QVPL shall be prepared by the contractor for all subassemblies and parts for approval by the procuring activity. All entries on the AS/QVPL shall be fully substantiated by the preparation of data to satisfy the requirements of source approval and part-test approval as specified herein. Reports of all quality verification testing shall conform to the requirements of MIL-STD-831. The contractor may solicit from the procuring activity information as to the status of quality-verified parts and approved sources which may have been developed previously. Neither part environmental tests nor part environmental test reports are required in those instances where vendors are selected from current AS/QVPLs of other programs. Entries on the AS/QVPL shall identify the vendor and the product which is proposed by the contractor for use in the generator. In the event several sources have been approved previously for the application, the contractor shall specify the source of the parts to be used. Each item listed shall be supported in accordance with the provisions of this section. This listing does not supplant or modify the applicable documentation or other Government Qualified Parts Listing.

3.2.4.1 Conditions governing the quality verification of parts. When the contractor proposes to demonstrate the quality verification of parts, as provided herein, the following provisions shall apply:

a. Test may be performed by the contractor, utilizing his own facilities, the facility of the manufacturer of the part or by any other suitable facility selected by the contractor. In any event, the facilities shall be approved by the contractor and the procuring activity prior to the performance of any test.

b. Prior to the performance of any test, a detailed test procedure shall be prepared by the contractor and approved by the procuring activity. The procedure shall identify, as a minimum, the step by step test procedures to be performed and the test equipment to be used, and shall provide for full conformance to the provisions of the applicable documentation. The Government reserves the right to review any of the individual test procedures, at its discretion.

c. Quality verification testing may be monitored by a representative of the procuring activity.

3.2.4.2 Changes to the AS/QVPL. The contractor may develop alternative sources for any part, subject to the provisions herein. Proposals by the contractor for part substitution or changes to the AS/QVPL shall include proof that the proposed substitute part will meet the requirements of 3.2.2 and 3.2.3. The AS/QVPL shall be revised and updated, as applicable, to include approved changes.

3.2.4.3 Removal from the AS/QVPL. Evidence of poor quality may be cause for suspension or removal of a source from the AS/QVPL at the option of the contractor or the procuring activity. The contractor shall develop quality verified sources for those parts for which the approved source has been removed. Quality reverification shall be required under any of the following conditions, unless otherwise directed by the procuring activity:

a. When the part has been modified by a change in design, materials, manufacturing process, manufacturing inspection procedures, or manufacturing facilities. The procuring activity shall be notified prior to delivery of any parts incorporating any such changes, and be provided quantitative evidence of the effect of such changes on the performance or other characteristics of the generator.

b. When there is evidence that the quality of the part has not been maintained. This evidence may be in the form of accumulated failure reports which indicate that the quality of the part has deteriorated, or there is a failure trend.

c. When the manufacturer has not produced the item for six or more months.

3.2.5 Part conformance provisions. When a drawing or specification for a part invokes by reference the provisions of a military specification, total conformance to the provisions of the referenced specification shall be required. Total conformance is defined as conformance to the stated performance and product characteristics, and compliance with all quality assurance provisions, including qualification, quality verification, and quality conformance testing.

3.3 Design and construction. The design and construction of the generator shall be in accordance with the applicable drawings listed on DL 1236AS500, and as specified herein. In the event of conflict between the drawings listed on DL 1236AS500 and this specification, the requirements of this specification shall apply. The contractor shall notify the procuring activity in accordance with DOD-STD-480, of any changes in design and construction after the generator has been accepted by the procuring activity.

3.3.1 Materials. Materials shall be selected on the basis of strength, workability and compatibility with other materials in the item so as to minimize galvanic action, corrosion, etc.

3.3.2 Finishes. Finishes shall be applied to all materials, whenever applicable, and shall be for the purpose intended, such as corrosion resistance, increased durability, etc. External surfaces shall be in accordance with the applicable drawings listed on DL 1236AS500.

3.3.3 Interchangeability. All generators having the same part number shall be internally and externally identical. Changes in part numbers shall be governed by drawing number requirements of DOD-STD-100.

3.3.4 Identification of parts. All generators shall be marked as specified on the applicable drawings listed on DL 123AS500 or in accordance with MIL-STD-130, if not specified on the drawing.

3.3.5 Dimensions. The generator envelope dimensions shall be in accordance with the applicable drawings listed on DL 1236AS500.

3.3.6 Weight. The weight of the generator shall be not greater than 5.0 pounds.

3.4 Performance. The generator, constructed in accordance with the applicable drawings listed on DL 1236AS500, shall meet the following requirements

3.4.1 Service conditions (electrical).

3.4.1.1 Primary input power. The generator shall give specified performance from a power source of 28 ±2.8 VDC with characteristics as defined in MIL-STD-704 for Category "B" equipment. The maximum current requirement shall be 2.0 amperes. Presence of DC voltage at the input power point shall cause the 28 VDC Status Indicator to illuminate (See Figure 1).

3.4.1.2 Warm-up time. The maximum warm-up time for the generator to generate a stable PRF (within ±1 pulse per second of the selected PRF) shall be not greater than five minutes for normal or extreme service conditions.

3.4.1.3 Undervoltage protection. The generator shall not be damaged when operated from a power source voltage below the minimum specified herein, and shall resume normal operation automatically when the voltage returns to 28 ±2.8 VDC.

3.4.1.4 Reverse polarity. The generator shall not be damaged by extended periods of application of reverse polarity voltage up to 30.8 VDC.

3.4.1.5 Electronic output and controls. The generator shall provide the following electronic output parameters under all specified environmental operating conditions:

a. Pulse width. Pulse width shall be 200 ±10 microseconds (see Figure 2a).

b. Frequency range. The generator shall generate from 3 to 1999 pulses per second up to the maximum usable pulse repetition frequency (PRF) as defined in Figure 3. An internal jumper switch shall allow the selection of a range from 3 to 3999 pulses per second up to the maximum usable frequency in Figure 3. (The PRF Hi Limit Jumper is located on the PRF Rate, Width, and Delay Circuit Card Assembly.)

c. Selectability. The generator shall be capable of selecting any frequency within the specified range within one pulse per second. The PRF Parameter Rate Thumbwheel Switches shall perform the selection function (see Figure 1).

d. Stability. The generator shall have stable mean frequency within ±1 pulse per second of the selected PRF Parameter up to the maximum usable PRF as defined in Figure 3 for all positions of the PRF Mode Switch.

e. Pulse amplitude. Pulse amplitudes shall be 12 ±0.5 VDC (see Figure 2a).

f. Quiescence level. The quiescence level (voltage during the interpulse period) shall be 0.15 ±0.15 VDC (see Figure 2a).

g. Jitter mode. The generator shall produce a random PRF jitter, which is internally adjustable from 10 to 50 microseconds peak-to-peak around the base PRF, when the "JITTER" mode is selected on the PRF Mode Switch (see Figure 1). (The potentiometer R2 on the PRF, Mode, Jitter and Output Circuit Card Assembly, adjusts the jitter.)

h. Stagger mode. The generator shall have an internally adjustable stagger interval (adjusted by potentiometer R1 on the PRF, Mode, Jitter and Output Circuit Card Assembly) of two to ten microseconds. The generator shall have a stagger mode selection capability of STAG 4, STAG 8, and STAG 16 on the PRF Mode Switch (see Figure 1). These stagger modes shall cause a steadily increasing interval (increasing by two to ten microseconds determined by the setting of the stagger interval) between pulses for up to 3, 7, or 15 pulses in the series depending upon the STAG 4, STAG 8, or STAG 16 setting. The respective 4th, 8th, or 16th pulse shall follow a shorter interval which corrects for the sum of the previous stagger displacements in that cycle. The result is that for a stagger mode of n, the 1, n+1, 2n+1, ... pulses in the pulse train shall occur at the same time that the pulses would have occurred if no stagger had been introduced. This stagger mode is defined more completely in Figure 4 for a setting of STAG 4 on the PRF Mode Switch. A STAG 8 or a STAG 16 setting shall produce similar pulse trains, except the quiescent interval shall continue to increase until after the seventh pulse for STAG 8 or the fifteenth pulse for STAG 16. The correction quiescent interval shall then occur and the series shall be repeated. The staggered pulse mode shall be operational (when selected) for all settings of the Scan Mode Switch (see Figure 1).

i. Electronic output indication. The generator shall illuminate the Radiate Status Indicator when it receives a ground from the associated simulating set pod through pin C of the input/output connector.

j. Fault status indicator. A selected delay setting that is not less than the rate setting minus the width setting shall cause the Fault Status Indicator light to illuminate (see Figure 1).

3.4.2 Scan parameters.

3.4.2.1 Rate. Rate shall be adjustable in 0.1 second increments between 0.1 seconds and 9.9 seconds by movement of the two Scan Parameter Rate Thumbwheel Switches labeled "RATE" (see Figure 1). Rate shall be identified separately for the available scan modes according to the following:

a. Circular scan mode rate. When operating in the circular scan mode ("CIRC" selected on the Scan Mode Switch as shown in Figure 1) rate shall be the time, in seconds, between the leading edges of adjacent groups of pulses (see Figure 2a).

b. Sector scan mode rate. When operating in either horizontal sector scan mode ("HS CTR" on the Scan Mode Switch as shown in Figure 1) or the horizontal sector off-center mode ("HS OFF CTR" on the Scan Mode Switch as shown in Figure 1) rate shall be the time between every other group of pulses (see Figure 2b).

3.4.2.2 Width. Width, which is the length of time in seconds between the rise of the first pulse in a pulse group and the fall of the last pulse in that pulse group (see Figure 2a), shall be adjustable in 0.1 second increments between 0.1 seconds and 0.9 seconds by movement of the Scan Parameter Width Thumbwheel labeled "WIDTH" (see Figure 1).

3.4.2.3 Delay. Delay, which is the length of time in seconds that every other pulse group is delayed (see Figure 2c), is used to simulate the off-center scanning search period of a radar. The delay function shall be activated only when the horizontal sector off-center mode ("HS OFF CTR") is selected (see Figure 1). The delay shall be adjustable in 0.1 second increments between 0.1 seconds and 3.9 seconds by movement of the two Scan Parameter Delay Thumbwheel Switches labeled "DELAY" (see Figure 1).

3.4.2.4 Steady scan mode. The selection of the Steady Scan Mode (Scan Mode Switch in "STEADY" position as shown in Figure 1) shall cause a continuous train of pulses to be transmitted at the PRF selected.

3.4.3 Electromagnetic radiation. The generator shall meet the electromagnetic interference requirements of MIL-STD-461 for Category Alb equipment.

3.4.4 Service conditions (environmental). The generator shall operate satisfactorily under any of the environmental service conditions or combination of these conditions as specified herein.

3.4.4.1 Low pressure (altitude). The generator shall operate satisfactorily at altitudes up to and including 50,000 feet above mean sea level.

3.4.4.2 High temperature. The generator shall operate satisfactorily during and after the test of MIL-STD-810, Method 501.1, Procedure I using a maximum operating temperature of 71°C. The generator shall operate satisfactorily after a period of four (4) hours of storage at 95°C.

3.4.4.3 Low temperature. The generator shall operate satisfactorily after the test of MIL-STD-810, Method 502.1 using a storage temperature of -62°C for a period of four (4) hours and a lowest operating temperature of -54°C.

3.4.4.4 Thermal shock. The generator shall operate satisfactorily after the test of MIL-STD-810, Method 503.1, Procedure I.

3.4.4.5 Humidity. The generator shall operate satisfactorily after the test of MIL-STD-810, Method 507.1, Procedure I, except that four 24 hour cycles shall be substituted for the ten 24 hour cycles.

3.4.4.6 Vibration. The generator shall operate satisfactorily during and after the test of MIL-STD-810, Method 514.2, using vibration curve J for equipment category b.1.

3.4.4.7 Shock. The generator shall operate when subjected to the shock requirements of MIL-T-5422 except that the shock level shall be 20 Gs for the Basic Design Test. Specified performance is required only after shock.

3.4.5 Service life. The generator shall operate satisfactorily and show no damage during and after being subjected to the service life test as specified herein.

·ship displayed in fabrication and assembly of
th·o assure, within design limitations, ability
of formance requirements under all applicable
en·ed herein. Particular attention shall be
pa·s of assembly, marking of parts and assem-
bl·ing, welding, brazing and freedom of parts
fr· standards of workmanship exhibited in the
ap·ubject to any qualification stated in the
go· shall determine the requirements of the
re·ve to workmanship.

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4tion. Unless otherwise specified in the con-
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req Except as otherwise specified in the con-
tra· own or any other facilities suitable for
the· requirements specified herein, unless dis-
app Government reserves the right to perform any
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4.on. The inspection of the generators shall
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ıspection (see 4.5).

4ıless otherwise specified herein, all inspec-
tio·e following conditions and tolerances.

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parts of the test item shall be stabilized
erature ±2.8°C (±5°F) indicated prior to con-
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4.4 First article inspection. First article inspection shall consist of the tests outlined in Table I. The procuring activity reserves the right to delete tests and alter the test sequence of Table I (see 6.2.1). First article approval is valid only on the contract or purchase order under which it is granted, unless extended by the Government to other contracts or purchase orders.

4.4.1 Generator samples. Unless otherwise specified in the contract, first article generator samples shall consist of the first two generators manufactured on the contract or purchase order. The generator shall be labeled permanently for ease of identification and shall be produced with equipment and procedures that will be used for the production of all the generators designated in the contract or purchase order. Prior to submission, the contractor shall inspect the samples to the degree necessary to assure that they conform to the requirements of the contract and prepare a first article test report of this inspection with the samples, including statements of findings and certificates of conformance for materials, and tests A through P of Table I.

4.5 Quality conformance inspection. Acceptance of generators shall be on an inspection-lot basis. Quality conformance inspection shall consist of the following tests:

a. Individual tests (see 4.5.1).

b. Sampling tests (see 4.5.2).

4.5.1 Individual tests. Individual tests are those tests conducted by the contractor on each generator submitted for acceptance under the contract or purchase order. Individual tests for generators are shown in Table II, Group 1. Failure to pass these tests shall be cause for rejection.

4.5.2 Sampling tests. A random sampling of generators shall be selected by the procuring activity from each inspection lot in accordance with Table III and shall be subjected to the tests listed in Table II, Groups 1 and 2. Group 1 tests shall be conducted prior to Group 2 tests. Failure to pass these tests shall be cause for rejection of the lot. The procuring activity reserves the right to delete tests and alter the test sequence of Table II (see 6.2.1). In the event any sample representing an inspection lot fails to meet the inspection and test requirements outlined herein, the contractor, at the discretion of the procuring activity, may have the opportunity of reworking the entire affected lot (see 6.2.1). The procuring activity shall determine which inspections and tests of this specification are to be performed on the reworked inspection-lot samples (see 6.2.1). The number of test samples shall be at the discretion of the procuring activity (see 6.2.1).

TABLE I. First article inspections - schedule and sequence.[1/] [2/]

Test	Test title	Requirement paragraph	Test method paragraph	Sample	
				A	B
A	Inspection	3.3	4.6.1	1	1
B	Primary input power	3.4.1.1	4.6.2	2	2
C	Warm-up time	3.4.1.2	4.6.3	3	3
D	Undervoltage protection	3.4.1.3	4.6.4	4	4
E	Reverse polarity	3.4.1.4	4.6.5	5	5
F	Electronic output and controls	3.4.1.5	4.6.6	6	6
G	Scan parameters	3.4.2	4.6.7	7	7
H	Electromagnetic radiation	3.4.3	4.6.8	8	X
I	Low pressure	3.4.4.1	4.6.9.1	9	X
J	High temperature	3.4.4.2	4.6.9.2	10	X
K	Low temperature	3.4.4.3	4.6.9.3	11	X
L	Thermal shock	3.4.4.4	4.6.9.4	12	X
M	Humidity	3.4.4.5	4.6.9.5	13	X
N	Vibration	3.4.4.6	4.6.9.6	14	X
O	Shock	3.4.4.7	4.6.9.7	15	X
P	Service life	3.4.5	4.6.10	X	8

[1/] Test sequence is denoted by the numbers in the sample columns. An X denotes test is not required.

[2/] If only one item is submitted for testing the service life test shall follow the low pressure test (moving tests J through O to follow it in the sequence).

TABLE II. Quality conformance inspections - schedule and sequence.[1]

Tests	Requirement paragraph	Test method paragraph	Sample			
GROUP 1			ALL SAMPLES			
Inspection	3.3	4.6.1	1			
Primary input power	3.4.1.1	4.6.2	2			
Warmup time	3.4.1.2	4.6.3	3			
Undervoltage protection	3.4.1.3	4.6.4	4			
Reverse polarity	3.4.1.4	4.6.5	5			
Electronic output and controls	3.4.1.5	4.6.6	6			
Scan parameters	3.4.2	4.6.7	7			
GROUP 2[2]			A	B	C	D
Electromagnetic radiation	3.4.3	4.6.8	1	X	X	X
Low pressure	3.4.4.1	4.6.9.1	2	X	1	X
High temperature	3.4.4.2	4.6.9.2	3	X	2	X
Low temperature	3.4.4.3	4.6.9.3	4	X	3	X
Thermal shock	3.4.4.4	4.6.9.4	5	X	4	X
Humidity	3.4.4.5	4.6.9.5	6	X	5	X
Vibration	3.4.4.6	4.6.9.6	7	X	6	X
Shock	3.4.4.7	4.6.9.7	8	X	7	X
Service life	3.4.5	4.6.10	X	3	X	1

[1] Test sequence is denoted by the numbers in the sample columns. An X denotes the test is not required.

[2] If contract specifies a sample of one unit, the service life test shall be accomplished ahead of the other Group 2 tests and the remaining Group 2 tests shall then be accomplished.

TABLE III. Sampling test items.

Lot size	Number of samples to be tested
100 or less	2
101 to 300	3
301 to 500	4
501 or more	4 plus 1 for each additional 500 or fraction thereof.

4.6 Inspection methods.

4.6.1 Inspection. The generator shall be inspected as follows.

4.6.1.1 Individual visual tests. The generator shall be inspected to ensure that it has been properly assembled, and the workmanship is as specified in 3.5.

4.6.1.2 First article and quality conformance tests. The generator shall be subjected to the examinations and tests of Tables I and II. Failure of the item to pass examinations and tests within the criteria specified in 4.5.2 shall cause rejection of the lot. The generator shall be inspected as follows:

a. The generator shall be thoroughly inspected to determine conformance to all the requirements of the applicable drawings listed on DL 1236AS500 and this specification.

b. The generator shall be examined to ascertain that the packaging, packing and marking is in accordance with Section 5.

4.6.2 Primary input power. For test purposes, the generator shall be provided a power source capable of adjustment between 0 VDC and 30.8 VDC. The application of 28 ±2.8 VDC input power shall cause the 28 VDC Status Indicator to illuminate (see Figure 1). The current requirement for the remaining tests will be monitored, and the maximum shall not exceed 2.0 amperes.

4.6.3 Warm-up time. The generator shall be connected to the primary input power source and the time shall be measured between the time of connection and the time a stable pulse train (within ±1 PRF) is generated. This test may be conducted in conjunction with the primary input power test 4.6.2 using ambient environmental conditions. For extreme temperatures (Group 2 tests only) it shall be conducted in conjunction with the high temperature test (see 4.6.9.2) and the low temperature test (see 4.6.9.3).

4.6.4 Undervoltage protection. The generator shall be placed in normal operation, and the input voltage reduced to 14 ±1 VDC while the generator is operating. The lower voltage shall be held for two minutes and then returned to 28 ±2.8 VDC. The generator shall then operate properly as determined by successful completion of the electronic output test (see 4.6.6).

4.6.5 Reverse polarity. The input voltage to the generator shall be increased to 30.8 VDC, and the polarity shall be reversed for a period of one minute. Specified input voltage (28 ±2.8 VDC) shall then be applied. The generator shall operate properly as determined by successful completion of the electronic output test (see 4.6.6).

4.6.6 Electronic output. The generator shall be placed in normal operation with the input voltage at 25.3 ±0.1 VDC. By use of an oscilloscope and a frequency counter, the pulse parameters (pulse width, pulse repetition frequency (PRF), pulse stability, pulse amplitude, quiescent level, and the interpulse intervals when in the stagger mode) shall be measured using test points 1 through 7, and 10 and 12 of Table IV. For test point 10 the interval jitter potentiometer (R2) will be set at minimum and the jitter shall be not greater than 10 microseconds peak-to-peak around the base PRF. For test point 12 the interval jitter potentiometer (R2) shall be set at maximum and the jitter shall be no less than 50 microseconds peak-to-peak around the base PRF. Any failure of the test item to provide the electronic output parameters (see 3.4.1.5) shall cause rejection of the test item.

4.6.7 Scan parameters. The generator shall be placed in normal operation with the input voltage at 30.7 ±0.1 VDC. An oscilloscope shall be used to measure the scan parameters (rate, width, and delay) for test points 8, 9, 11, 13, and 14. Any failure of the test item to provide the scan parameters (see 3.4.3) shall cause rejection of the test item.

4.6.8 Electromagnetic radiation. The generator shall be subjected to the tests of Table V in accordance with MIL-STD-462. Specified performance shall include successful completion of the electronic output test (see 4.6.6) and the scan parameter test (see 4.6.7). The generator shall be rejected if it fails to meet the limits defined in MIL-STD-461.

4.6.9 Environmental tests. The generator shall be subjected to the following environmental tests.

4.6.9.1 Low pressure. The generator shall be subjected to the low pressure test in accordance with MIL-STD-810, Method 500.1 except a second operating altitude shall be tested at 50,000 feet above sea level. The generator shall be held at 50,000 feet for at least one hour and then operated by performing the electronic output test (see 4.6.6) and the scan parameter test (see 4.6.7). The generator shall be rejected if it does not meet the requirements of both the electronic output test and the scan parameter test during and after the low pressure test.

TABLE IV. Electronic output and scan parameters test points.[1]

Test[2] Points	Scan Switch	Mode[3] Position	Rad Status Indicator	Fault Status Indicator	PRF Switches	Rate Switches	Delay Switches	Width Switch	PRF Hi-Limit Jumper
1	Steady	Stable	on	off	0003	N/A	N/A	N/A	A
2	Steady	Stable	on	off	1999	N/A	N/A	N/A	A
3	Steady	Stable	on	off	1000	N/A	N/A	N/A	A
4	Steady	Stable	on	off	3999	N/A	N/A	N/A	B
5	Steady	Stag 4	on	off	1000	N/A	N/A	N/A	B
6	Steady	Stag 8	on	off	1000	N/A	N/A	N/A	B
7	Steady	Stag 16	on	off	1000	N/A	N/A	N/A	B
8	HS off Ctr	Stable	on	on	1000	97	89	9	B
9	Circ	Stable	on	off	1000	26	N/A	9	B
10	Circ	Jitter	on	off	3999	26	N/A	9	B
11	HS Ctr	Stable	on	off	1000	99	N/A	1	B
12	HS Ctr	Jitter	on	off	3999	99	N/A	1	B
13	HS Ctr	Stable	on	off	1000	50	N/A	5	B
14	HS off Ctr	Stable	on	off	1000	50	10	5	B

[1] N/A denotes that the setting has no effect on the test.

[2] The order of testing need not be ascending by test point number.

[3] The internal stagger interval shall be set at 5 ±1 microseconds.

TABLE V. Electromagnetic tests.

Test method	Description
CE01	30 Hz to 20 kHz, Power Leads
CE02	0.03 to 20 kHz, Control and Signal Leads
CS01	0.03 to 50 kHz, Power Leads
CS02	0.05 to 400 MHz, Power Leads
CS06	Spike, Power Leads
RE01	0.03 to 30 kHz, Magnetic Field
RS01	0.03 to 30 kHz, Magnetic Field
RS02	Magnetic Induction Field

4.6.9.2 High temperature. The generator shall be subjected to the high temperature test in accordance with MIL-STD-810, Method 501.1, Procedure I, using a maximum operating temperature of 71°C. The electronic output test (see 4.6.6) and the scan parameter test (see 4.6.7) shall be conducted at both the elevated temperature and at the completion of the test. The generator shall be rejected if it fails either the electronic output test or the scan parameter test.

4.6.9.3 Low temperature. The generator shall be subjected to the low temperature test in accordance with MIL-STD-810, Method 502.1, Procedure I for a period of four (4) hours. The storage temperature shall be -62°C and the lowest operating temperature shall be -54°C. The generator shall be operated by performing both the electronic output test (see 4.6.6) and the scan parameter test (see 4.6.7) during and after the low temperature test. The generator shall be rejected if it fails either the electronic output test or the scan parameter test.

4.6.9.4 Thermal shock. The generator shall be subjected to the thermal shock test in accordance with MIL-STD-810, Method 503.1, Procedure I. At the completion of the test, the generator shall be operated by performing both the electronic output test (see 4.6.6) and the scan parameter test (see 4.6.7). The generator shall be rejected if it fails either test.

4.6.9.5 Humidity. The generator shall be subjected to the humidity test in accordance with MIL-STD-810, Method 507.1, Procedure I, except that four 24-hour cycles shall be substituted for the ten 24-hour cycles. The equipment shall be operated after the test by conducting both the electronic output test (see 4.6.6) and the scan parameter test (see 4.6.7). The generator shall be rejected if it fails either test.

4.6.9.6 Vibration. The generator shall be subjected to the following tests.

4.6.9.6.1 General. The vibration tests specified herein shall be performed in each of three mutually perpendicular axes in ambient environmental conditions. All testing shall be performed in one axis prior to testing in the next axis. The generator shall be checked for proper operation, as specified herein.

4.6.9.6.2 Generator operation. When performance checks are specified in MIL-T-5422, Procedure I, they shall be accomplished using 4.6.6 and 4.6.7.

4.6.9.6.3 Mounting techniques. The generator shall be mounted in a rigid panel fabricated to simulate a typical instrument panel installation. The panel shall be attached to the vibration table. The vibration input shall be measured on the fixture at a point adjacent to the mounting point.

4.6.9.6.4 Sinusoidal vibration tests. The generator shall be subjected to sinusoidal vibration along each of three mutually perpendicular axes in accordance with MIL-T-5422, Procedure I using Curve Ia. The generator shall rejected if it fails any performance check specified in MIL-T-5422, Procedu I or if it shows any physical deformation during specified visual checks.

4.6.9.7 Shock. The generator shall be subjected to the shock test in accordance with MIL-T-5422, except that the Part II - Crash Safety Test shall not be performed. The peak value Gs shall be 20 Gs instead of 15 Gs. The equipment shall continue to operate during the test. Specified performance shall be required only after the test. Both the electronic test (see 4.6.6) and the scan parameter test (see 4.6.7) shall be conducted after the completion of the shock test. The generator shall be rejected if it fails either test or if it shows any evidence of mechanical failure.

4.6.10 Service life. The generator shall be subjected to reliability acceptance testing in accordance with MIL-STD-781 Test Plan XXIC using a lower test mean-time-between-failures of 50 hours. A typical mission profile for an air superiority fighter (Table B-II) and an airborne mission length of 1.25 hours shall be used. A reliability test plan document and reliability test procedures shall be prepared for procuring activity approval piror to initiating the test. Performance of the equipment during the test shall consist of continuous cycling of the equipment using the electronic output test (see 4.6.6) and the scan parameter test (see 4.6.7). Failure actions shall be defined in the test plan. Any failure of the generator to perform as specified herein shall be classified as a failure.

5. PACKAGING

5.1 Generator preservation and packaging. Generators shall be preserved and packed in accordance with MIL-STD-794 using Level C Criteria.

5.2 Generator container marking. All unit packages and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use. The Generator-Control-Indicator SG-1189/AST-4(V) is intended for use in remote control of Transmitting Set, Radar AN/DPT-1(V) from the cockpits of various types of aircraft. The AN/DPT-1(V) is mounted in a Pod, Simulating Set CY-7927/AST-4(V), which is an airborne, multimode radar transmitter (pod mounted) that simulates the pulse-modulated microwave carriers of certain types of radar transmitters.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

a. Title, number, revision letter, and date of this specification.

b. First article requirements (see 3.1, 3.1.2, and 4.4.1).

c. Designated testing activity (see 3.1.2).

d. Changes to the AS/QVPL (see 3.2.4.2 and 3.2.4.3).

e. Option for quality reverification (see 3.2.4.3).

f. Responsibility for inspection and inspection conditions (see 4.1 and 4.3).

g. Option for government to perform any inspection (see 4.1).

h. Samples subjected to sampling inspection (see 4.5.2) shall not be considered or accepted as part of the contract or purchase order.

i. Rework and retest requirements and quantity to be submitted for inspection lot samples (see 4.5.2).

j. Test deleted and altered test sequence (see 4.4 and 4.5.2).

6.2.2 Data requirements. When this specification is used in a procurement which incorporates a DD Form 1423 and invokes the provisions of 7-104.9(n) of the Defense Acquisition Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of DAR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

Paragraph	Data requirement	Applicable DID
3.2	Test Reports	DI-T-21466A
3.2.1, 3.2.2 3.2.4 & 3.2.4.2	Approved Parts Source	UDI-R-20331B
3.2.4.1	Test Procedures	DI-T-3702A
3.3	Engineering Change Proposal	DI-E-2037
4.4.1	Report, First Article Test	UDI-T-23790
4.6.10	Plan, Reliability Test	DI-R-7033
4.6.10	Procedures, Reliability Tests	DI-R-7035

(Copies of data item descriptions required by the contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.3 First article. When a first article is required, it shall be tested and approved under the appropriate provisions of 7-104.55 of the Defense Acquisition Regulation. The first article should be a first production item. The first article should consist of two generators. The contracting officer should include specific instructions in all procurement instruments, regarding arrangements for examinations, tests, approval and disposition of the first article.

6.4 Contract or purchase order provision. Contracts or purchase orders should specify the following provision for first article inspection. "When a contractor is in continuous production of the generators applicable to this specification from contract to contract, consideration shall be given to waive the first article inspection. However, waiver of first article inspection shall not relieve the contractor of his obligation to fulfill all other requirements of the specification and contract." The words "purchase order" should be substituted throughout for the word "contract" when applicable.

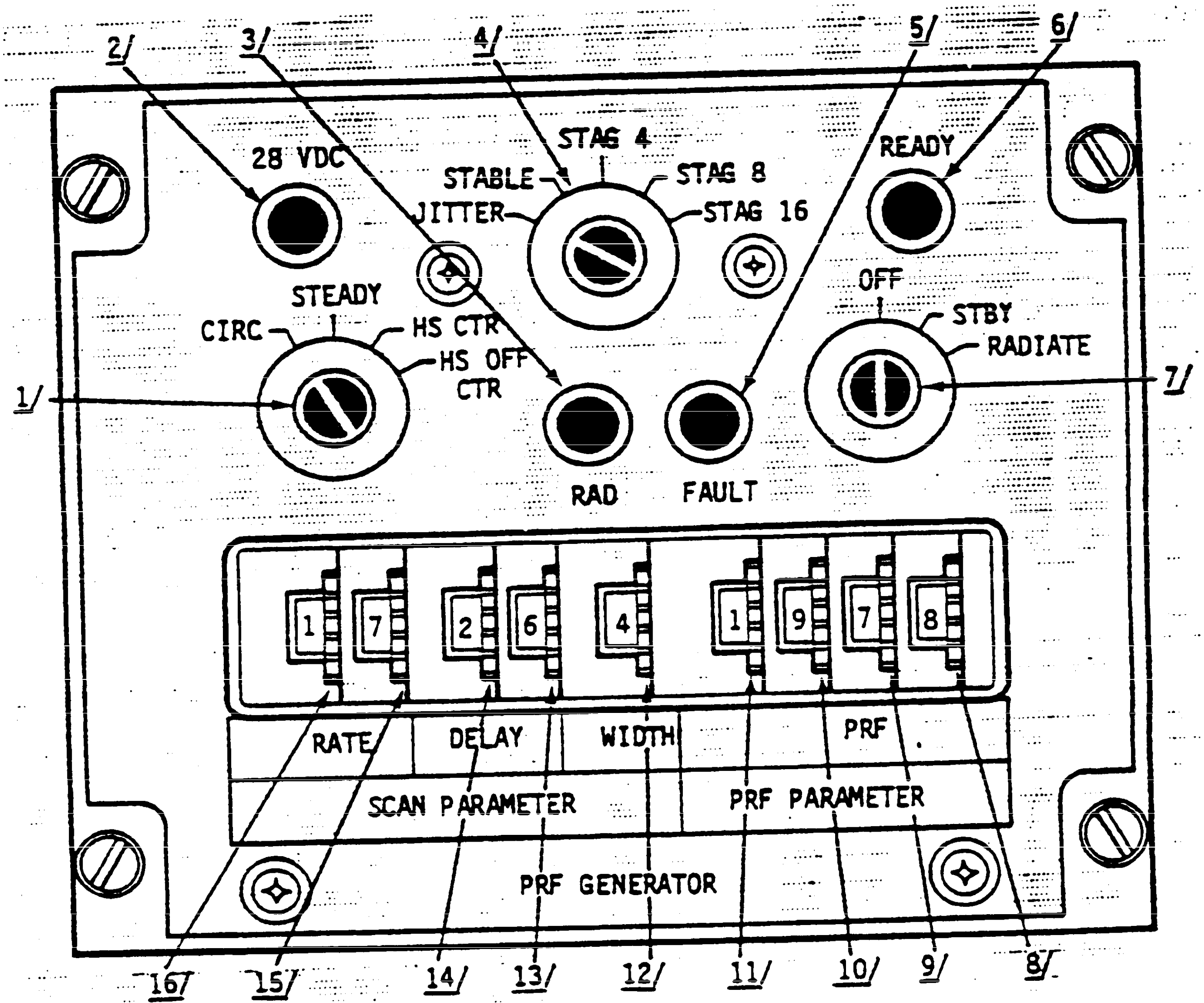
6.5 Precedence. The selection of specifications and standards for use in the design and development should be in accordance with the requirements of MIL-STD-143. When the requirements, this specification, and referenced documents are in conflict, the following precedence should apply:

a. This specification should have precedence over all referenced documents.

b. Any documents referenced in this specification should have precedence over all subsidiary documents referenced therein.

Preparing activity:
Navy - AS

(Project 5841-N018)


2/
3/
4/
5/
6/
28 VDC
STAG 4
STABLE
STAG 8
JITTER
STAG 16
READY
STEADY
CIRC
HS CTR
HS OFF CTR
OFF
STBY
RADIATE
1/
7/
RAD
FAULT
1
7
2
6
4
1
9
7
8
RATE
DELAY
WIDTH
PRF
SCAN PARAMETER
PRF PARAMETER
PRF GENERATOR
16/
15/
14/
13/
12/
11/
10/
9/
8/

1/ Scan Mode Switch
2/ 28 VDC Status Indicator
3/ Radiate Status Indicator
4/ PRF Mode Switch
5/ Fault Status Indicator
6/ Ready Status Indicator
7/ Control Switch
8/, 9/, 10/, and 11/ PRF Parameter Rate Thumbwheel Switches
12/ Scan Parameter Width Thumbwheel Switch
13/ and 14/ Scan Parameter Delay Thumbwheel Switches
15/ and 16/ Scan Parameter Rate Thumbwheel Switches

FIGURE 1. Operator controls.

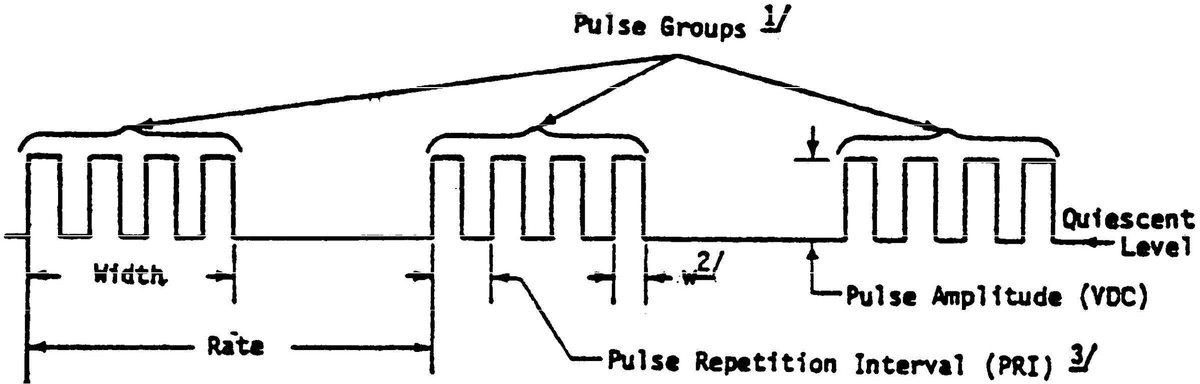

Pulse Groups 1/
Quiescent Level
Width
w 2/
Pulse Amplitude (VDC)
Rate
Pulse Repetition Interval (PRI) 3/

FIGURE 2a. Circular scan mode.

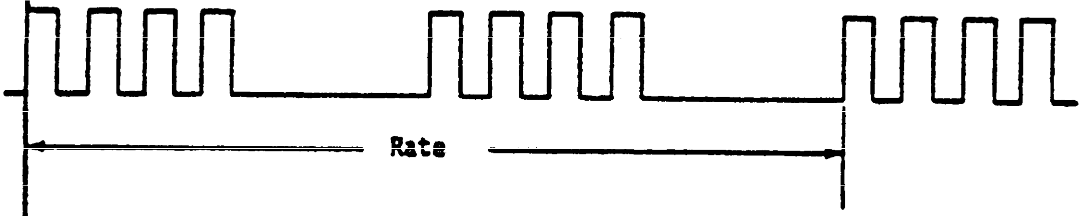


FIGURE 2b. Horizontal sector scan mode.

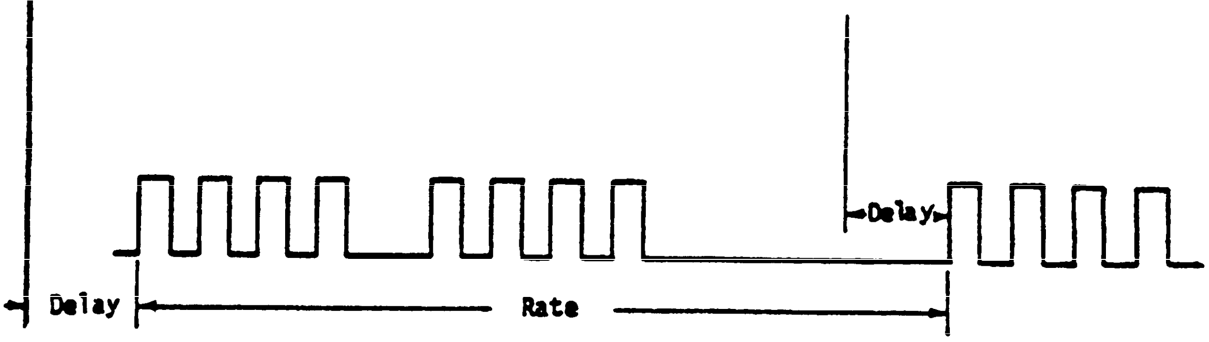

Delay
Delay
Rate

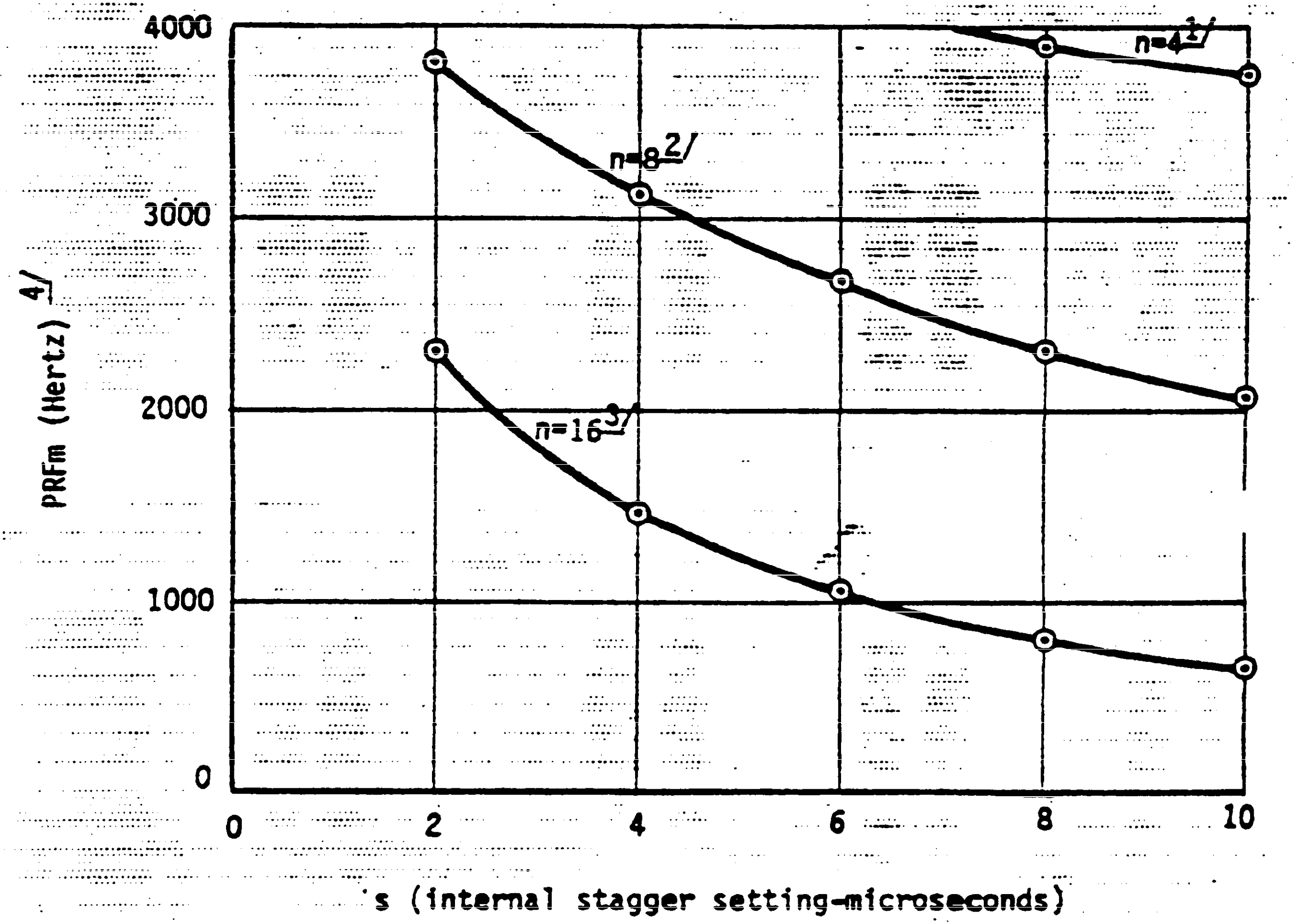
FIGURE 2c. Horizontal sector off center mode.

1/ Only four pulses in each pulse group are shown for ease in portrayal. The number in the group depends upon the pulse group width and PRF selected.

2/ w is pulse width in microseconds.

3/ The PRI in microseconds is equal to 10^6/PRF.

FIGURE 2. Scan mode parameters.



1/ PRF Mode Switch set at STAG 4.
2/ PRF Mode Switch set at STAG 8.
3/ PRF Mode Switch set at STAG 16.
4/ The maximum usable PRF (PRFm) can be calculated

by $\text{PRFm} = \dfrac{2 \times 10^6}{n(n-1)s+2w} \leq 3999$

where n, s, and w are as defined previously.
(PRFm without stagger is 3999 pulses per second.)

FIGURE 3. Maximum usable PRF (PRFm).

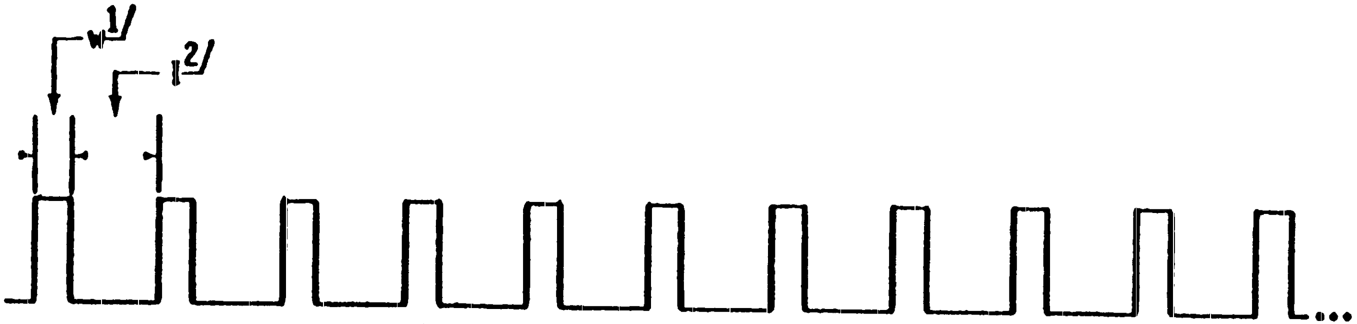


FIGURE 4a. Sample pulse train without stagger.

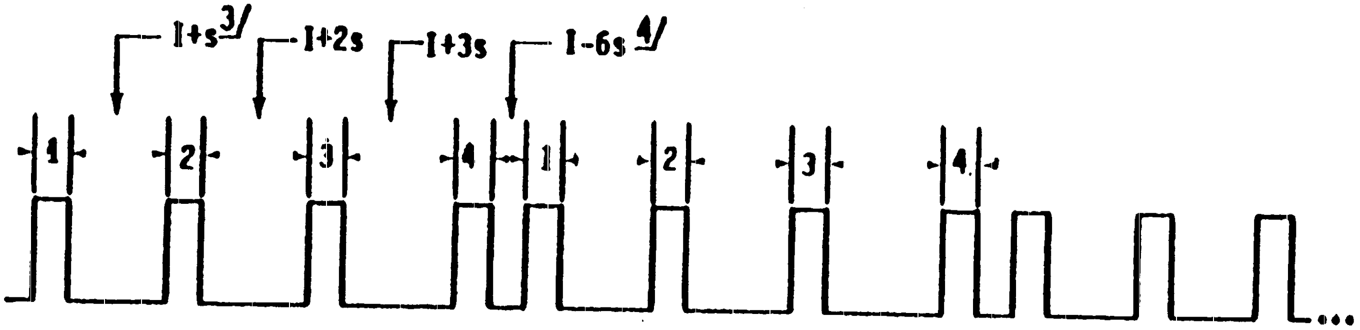


Figure 4b. Sample pulse train with stagger of 4 (STAG 4).

1/ w = pulse width of 200 ±10 microseconds

2/ I is defined as the quiescent interval between pulses without stagger selected.

3/ s is defined as the internally adjustable stagger interval which shall be from 2 to 10 microseconds.

4/ For stagger level setting of n, (for STAG 4, n = 4; STAG 8, n = 8; STAG 16, n = 16), the first pulse in each series shall follow a correction quiescent interval that forces the first pulse to occur at the same time that it would occur in a pulse train without stagger. (Compare Figure 4a with Figure 4b.) For this example, using a stagger level of STAG 4 (four pulses in a series before it repeats) and an internal stagger interval of s, the correction quiescent interval shall be I-6s microseconds.

The general equation for the correction quiescent interval (C) shall be $C = I - \frac{n(n-1)s}{2}$ where I, n, and s are as previously defined.

FIGURE 4. Comparison of pulse trains with and without stagger.

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